Communication Systems Lecture-13: Bandpass Signals

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Complex Envelope Representation (1)

Theorem: Any physical bandpass waveform (about $\pm f_c$) can be represented by:

$$v(t) = \mathrm{Re}\left\{g(t)e^{j\omega_c t}\right\}$$

- $\mathrm{Re}\{.\}$ stands for the real part of $\{.\}$.
- f_c is the associated carrier frequency ($\omega_c = 2\pi f_c$).
- $g(t)$ is called the complex envelope of $v(t)$.

Two other equivalent representations are:

$$\begin{aligned} v(t) &= R(t)\cos\left[\omega_c t + \theta(t)\right] \\ &= x(t)\cos(\omega_c t) - y(t)\sin(\omega_c t) \end{aligned}$$

where: $g(t) = x(t) + jy(t) = |g(t)|e^{j\arg[g(t)]} \triangleq R(t)e^{j\theta(t)}$.

- $x(t)$ is the in-phase modulation associated with $v(t)$.
- $y(t)$ is the quadrature modulation associated with $v(t)$.
- $R(t)$ is the amplitude modulation on $v(t)$.
- $\theta(t)$ is the phase modulation on $v(t)$.

Complex Envelope Representation (2)

- Expressing $g(t)$ in polar form as $g(t) = R(t)e^{j\theta(t)}$:

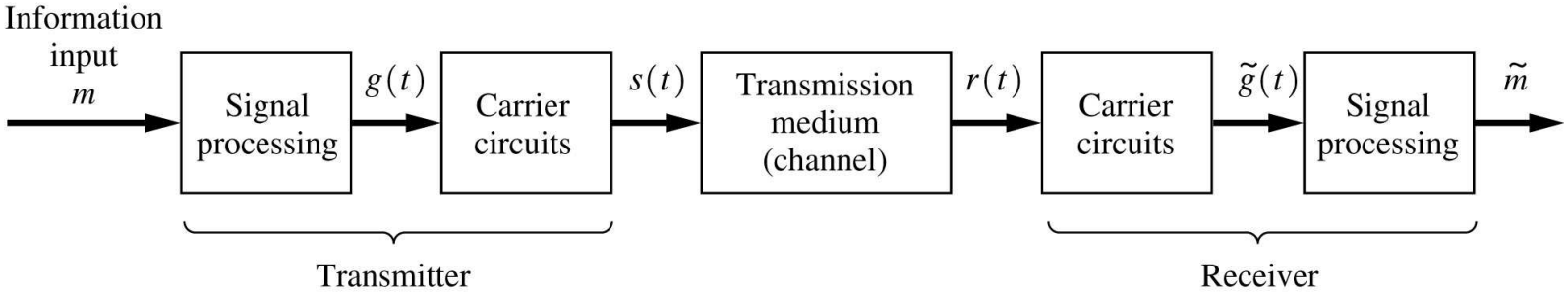
$$\begin{aligned} v(t) &= \mathrm{Re}\left\{g(t)e^{j\omega_c t}\right\} \\ &= \mathrm{Re}\left\{R(t)e^{j(\omega_c t+\theta(t))}\right\} = R(t)\cos\left[\omega_c t + \theta(t)\right] \end{aligned}$$

- In the same way, $g(t) = x(t) + jy(t)$ and:

$$\begin{aligned} v(t) &= \mathrm{Re}\left\{g(t)e^{j\omega_c t}\right\} \\ &= \mathrm{Re}\left\{\left[x(t) + jy(t)\right]\left[\cos(\omega_c t) + j\sin(\omega_c t)\right]\right\} \\ &= x(t)\cos(\omega_c t) - y(t)\sin(\omega_c t) \end{aligned}$$

- $v(t)$ (the modulated signal):
  - Bandpass and real.
- $g(t)$ (the complex envelope):
  - Baseband and complex.
- $x(t)$, $y(t)$, $R(t)$ and $\theta(t)$:
  - Baseband and real.

Representation of Modulated Signals (1)



- Modulation is the process of encoding the source information $m(t)$ (modulating signal) into a bandpass signal $s(t)$ (modulated signal).
- The modulated signal is given by:

$$s(t) = \mathrm{Re}\left\{g(t)e^{j\omega_c t}\right\}$$

where $\omega_c = 2\pi f_c$ and f_c is the carrier frequency.

- The complex envelope $g(t)$ is a function of $m(t)$:

$$g(t) = g\left[m(t)\right]$$

where $g[.]$ stands for a mapping operation.

Representation of Modulated Signals (2)

- Example: for Amplitude Modulation (AM) with $m(t) \geq -1$:

$$g(t) = A_c\,[1 + m(t)] = R(t)e^{j\theta(t)}$$
$$\Rightarrow \;\; R(t) = A_c\,[1 + m(t)] \quad ; \quad \theta(t) = 0$$

  and, consequently, the information is included only in the amplitude of the bandpass signal.

The mapping $g[m]$ is chosen to verify the following conditions:

- It must be easy to implement.
- It must give the desirable spectral properties.
- At the receiver, the inverse $m[g]$ should have a single value over the considered range.
- The mapping must suppress as much noise as possible.

Spectrum of Bandpass Signals (1)

Theorem: If a bandpass waveform is represented by:

$$v(t) = \mathrm{Re}\left\{g(t)e^{j\omega_c t}\right\}$$

then the spectrum of this waveform is:

$$V(f) = \frac{1}{2}\left[G(f - f_c) + G^*(-f - f_c)\right]$$

Proof:

- $v(t)$ can be written as:

$$\begin{aligned} v(t) &= \mathrm{Re}\left\{g(t)e^{j\omega_c t}\right\} \\ &= \frac{1}{2}g(t)e^{j\omega_c t} + \frac{1}{2}g^*(t)e^{-j\omega_c t} \end{aligned}$$

- Consequently, the Fourier transform of $v(t)$ is given by:

$$V(f) = \frac{1}{2}\mathcal{F}\left[g(t)e^{j\omega_c t}\right] + \frac{1}{2}\mathcal{F}\left[g^*(t)e^{-j\omega_c t}\right]$$

Spectrum of Bandpass Signals (2)

- Following from the frequency shifting property of the Fourier transform:

$$V(f) = \frac{1}{2}\left[\mathcal{F}\left[g(t)\right]\right]_{f-f_c} + \frac{1}{2}\left[\mathcal{F}\left[g^*(t)\right]\right]_{f+f_c}$$

- Given that $\mathcal{F}[g^*(t)] = G^*(-f)$:

$$\begin{aligned} V(f) &= \frac{1}{2}G\left(f - f_c\right) + \frac{1}{2}G^*\left[-(f + f_c)\right] \\ &= \frac{1}{2}\left[G(f - f_c) + G^*(-f - f_c)\right] \end{aligned}$$

- Note that if $g(t)$ is real, $g(t) = g^*(t)$ and $\mathcal{F}[g^*(t)] = G^*(-f) = \mathcal{F}[g(t)] = G(f)$ and consequently:

$$V(f) = \frac{1}{2}\left[G(f - f_c) + G(f + f_c)\right]$$

PSD of Bandpass Signals (1)

Theorem: If a bandpass waveform is represented by:

$$v(t) = \mathrm{Re}\left\{g(t)e^{j\omega_c t}\right\}$$

then its power spectral density is:

$$\mathcal{P}_v(f) = \frac{1}{4}\left[\mathcal{P}_g(f - f_c) + \mathcal{P}_g(-f - f_c)\right]$$

where $\mathcal{P}_g(f)$ is the PSD of $g(t)$.

Proof:

- It can be shown that the autocorrelation function of $v(t)$ is given by:

$$R_v(\tau) = \frac{1}{2}\mathrm{Re}\left\{R_g(\tau)e^{j\omega_c \tau}\right\}$$

PSD of Bandpass Signals (2)

- The last expression is similar to the complex envelope representation of signals. Consequently, based on the previous theorem:

$$\mathcal{P}_v(f) = \mathcal{F}\left[R_v(\tau)\right] = \frac{1}{4}\left[\mathcal{P}_g(f - f_c) + \mathcal{P}_g^*(-f - f_c)\right]$$

  where $\mathcal{P}_g(f) = \mathcal{F}\left[R_g(\tau)\right]$ is the PSD of $g(t)$.
- On the other hand, given that the PSD is real, $\mathcal{P}_g^*(f) = \mathcal{P}_g(f)$ and consequently:

$$\mathcal{P}_v(f) = \frac{1}{4}\left[\mathcal{P}_g(f - f_c) + \mathcal{P}_g(-f - f_c)\right]$$

- When $g(t)$ is real, then the PSD $\mathcal{P}_g(f)$ will be an even function implying that the last equation can be written as:

$$\mathcal{P}_v(f) = \frac{1}{4}\left[\mathcal{P}_g(f - f_c) + \mathcal{P}_g(f + f_c)\right]$$

Evaluation of Power

The average power of a bandpass signal $v(t)$ can be calculated from:

$$P_v = \langle v^2(t) \rangle = \int_{-\infty}^{+\infty} \mathcal{P}_v(f) \mathrm{d}f = R_v(0) = \frac{1}{2} \langle |g(t)|^2 \rangle$$

Proof:

- We have previously proven the first 3 equalities. On the other hand:

$$\begin{aligned} R_v(\tau) &= \frac{1}{2} \mathrm{Re} \left\{ R_g(\tau) e^{j\omega_c \tau} \right\} \\ \Rightarrow \quad R_v(0) &= \frac{1}{2} \mathrm{Re} \left\{ R_g(0) \right\} \\ &= \frac{1}{2} \mathrm{Re} \left\{ \langle g^*(t) g(t-0) \rangle \right\} \\ &= \frac{1}{2} \mathrm{Re} \left\{ \langle |g(t)|^2 \rangle \right\} = \frac{1}{2} \langle |g(t)|^2 \rangle \end{aligned}$$

where the last equation follows since $|g(t)|$ is always real.

Peak Envelope Power

- The Peak Envelope Power (PEP) is the average power that would be obtained if $|g(t)|$ is held at its peak value.
- The PEP is given by:

$$P_{PEP} = \frac{1}{2}\left[\max|g(t)|\right]^2$$

- In fact, let $g_1(t) \triangleq \max|g(t)| \triangleq A_p$. Based on the definition of the PEP:

$$\begin{aligned} P_{PEP} &= \frac{1}{2}\langle|g_1(t)|^2\rangle = \frac{1}{2}\langle A_p^2\rangle \\ &= \frac{1}{2}A_p^2 \\ &= \frac{1}{2}\left[\max|g(t)|\right]^2 \end{aligned}$$

Example: Amplitude-Modulated Signal (1)

- Designate by $m(t)$ the information message. For Amplitude Modulation (AM), the signal at the output of the signal processor is given by:

$$g(t) = A_c\,[1 + m(t)]$$

- Consequently, the spectrum of $g(t)$ is given by:

$$G(f) = A_c\delta(f) + A_c M(f)$$

where $M(f) = \mathcal{F}[m(t)]$.

- The spectrum of the baseband modulated signal $s(t)$ is:

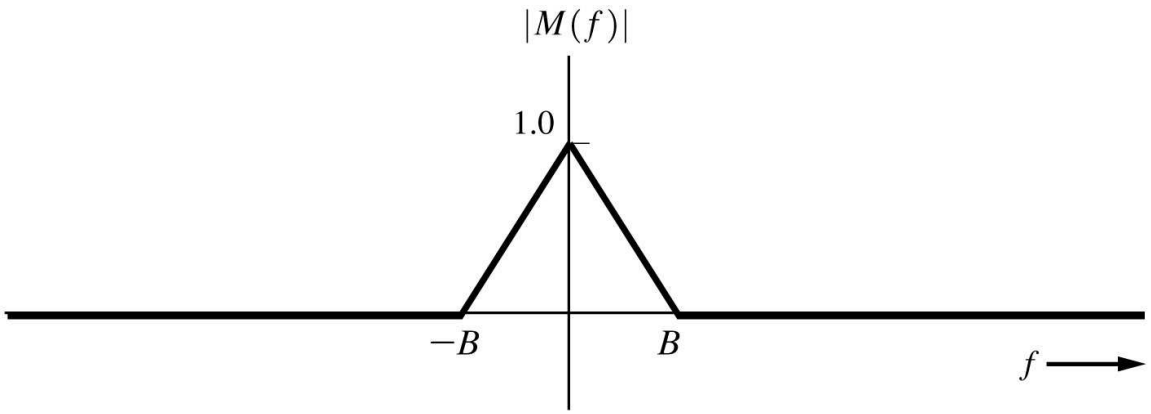
$$\begin{aligned} V(f) &= \frac{1}{2}\,[G(f - f_c) + G^*(-f - f_c)] \\ &= \frac{1}{2}\,[G(f - f_c) + G(f + f_c)] \end{aligned}$$

where the last equation follows since $m(t)$ is real implying that $g(t)$ is real and $G^*(f) = G(-f)$.

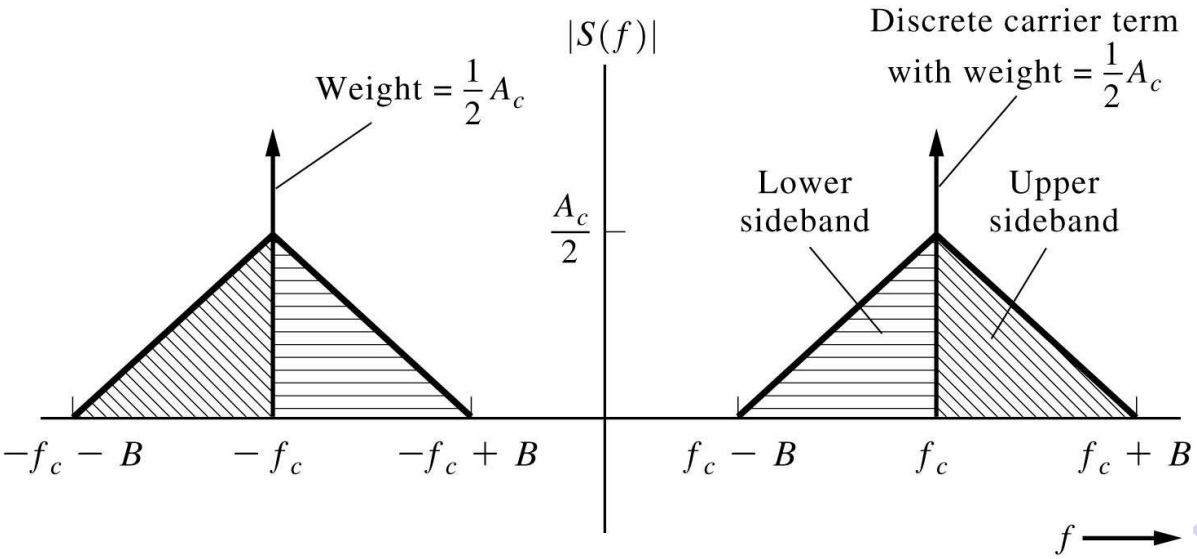
Example: Amplitude-Modulated Signal (2)

- Consequently:

$$S(f) = \frac{1}{2}A_c\left[\delta(f - f_c) + M(f - f_c) + \delta(f + f_c) + M(f + f_c)\right]$$



(a) Magnitude Spectrum of Modulation



Example: Amplitude-Modulated Signal (3)

- Note that $G(f - f_c)$ and $G(f + f_c)$ do not overlap since $f_c \gg B$ (where B is the absolute bandwidth of $m(t)$). Consequently, the magnitude spectrum is:

$$|S(f)| = \begin{cases} \frac{1}{2}A_c\delta(f - f_c) + \frac{1}{2}A_c|M(f - f_c)|, & f > 0; \\ \frac{1}{2}A_c\delta(f + f_c) + \frac{1}{2}A_c|M(f + f_c)|, & f < 0. \end{cases}$$

- The average power of the bandpass modulated signal $s(t)$ is given by:

$$\begin{aligned} P_s &= \frac{1}{2}\langle |g(t)|^2 \rangle \\ &= \frac{1}{2}A_c^2\langle |1 + m(t)|^2 \rangle \\ &= \frac{1}{2}A_c^2\langle 1 + 2m(t) + m^2(t) \rangle \\ &= \frac{1}{2}A_c^2\left[1 + 2\langle m(t) \rangle + \langle m^2(t) \rangle\right] \end{aligned}$$

Example: Amplitude-Modulated Signal (4)

- Assuming that the DC component of $m(t)$ is equal to zero: $\langle m(t) \rangle = 0$, then the average signal power becomes:

$$P_s = \frac{1}{2} A_c^2 \left[1 + P_m\right]$$

where: $P_m = \langle m^2(t) \rangle$ is the average power of the modulating signal $m(t)$.

  - $\frac{1}{2} A_c^2$ is the carrier power.
  - $\frac{1}{2} A_c^2 P_m$ is the power in the sidebands of $s(t)$.

- Note that since $m(t)$ is real, $s(t)$ can be written as:

$$\begin{aligned} s(t) &= \mathrm{Re}\left\{g(t) e^{j\omega_c t}\right\} \\ &= g(t) \cos(\omega_c t) \\ &= A_c \left[1 + m(t)\right] \cos(\omega_c t) \end{aligned}$$